

Diseases of Sugarcane

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Whip Smut of Sugarcane

The pathogen is *Ustilago scitaminea* Sydow, class Basidiomycetes, order Ustilaginales, family Ustilaginaceae.

Distribution

This disease has been reported from several sugarcane growing countries such as West Indies, Italy, Cuba, Mauritius, Java, Philippines, South Africa, and Australia. In India, the disease is found chiefly in the north, but occurs practically wherever cane is cultivated.

Symptoms

Affected plants are characterized by the production, from the growing apex, of a long whip-like dusty black shoot, often several feet in length and much curved on itself. The whip-like part of the cane is devoid of leaves, slender and flexible, and generally represents a floral shoot. In the earlier stages, the whip remains covered by a silvery-white, thin membrane, which soon ruptures exposing a dense black dust consisting of the spores (smut spores) of the fungus. No secondary shoots arise from the upper portion of the affected cane; however, from the lower part of the cane they are often fairly abundant, and may be prolonged into spore bearing organs. Generally the whip is found in those varieties which do not flower. Smut whips may arise from the upper part of cane or arise laterally.

The Pathogen

The whip smut of sugarcane is caused by *Ustilago scitaminea* Sydow. The tissues of the cane below the whip smut itself contain the intercellular mycelium. The hyphae are found in dense masses towards the surface of the spore-bearing shoot where the spores are formed. As the spores are produced in great abundance under the sheath, they rupture it, throwing it off in flakes and shreds. The spores are spherical, echinulate, light brown in colour, measuring 6.5 to 8.5 microns in diameter. The smut spores are intermingled with groups of colourless, thin-walled sterile cells. The spores are disseminated by wind and they readily germinate in water or moist earth, producing a two- or three-celled promycelium from which terminal and lateral sporidia arise. The sporidia are narrow, elongated, straight, single-celled bodies joined to the promycelium by a short stalk.

Nature and Recurrence of Disease

The disease is seed- as well as soil-borne. It can be transmitted through setts cut from diseased plants containing mycelium, as well as through spores that are in the soil. The mycelium of the fungus survives in the ratooned stumps of diseased canes, and fresh shoots which arise from such stumps are invariably smutted.

Control Measures

Rogueing: the smutted whips should be collected carefully in sterilized bags, and burnt outside the field. The entire infected clump should be dug out and burnt.

Avoid ratooning: the ratooning of infected crop should always be discouraged.

Avoid sowing of susceptible varieties: the susceptible varieties of cane to this disease should not be grown. Co. 300, Co. 301, Co. 312, Co. 313, and Co. 513 have been proved susceptible for this disease; Co. 213, Co. 299, and Co. 290 are also susceptible varieties.

Treatment of setts before sowing: prior to sowing, the setts should be dipped for five minutes in Bordeaux mixture 2:2:50, so that the smut spores adhering to them may be destroyed.

Hot-water treatment: prior to sowing, the setts should be dipped in water at 50 to 52°C for 20 to 25 minutes, so that the setts become disease free.

Healthy setts: the setts for sowing should be taken only from the healthy crops.

Resistant varieties: the resistant varieties should be sown. Some resistant varieties are Co. 356, Co. 395, Co. 421, Co. 453, P.O.J. 7878, Cos. 76, and Cos. 109.

Grassy Shoot Disease of Sugarcane

The causal agent is Sugar Cane Grassy Shoot Virus.

Distribution

This disease was first observed in Maharashtra State (Vasudeva, R.S., 1956). Now it is a common disease and one of the serious problems of Maharashtra. The disease has also been recorded from Andhra Pradesh, Tamil Nadu, Bihar, Orissa, Uttar Pradesh, and the Punjab. It has also been recorded from other countries such as Thailand and Taiwan.

Symptoms

The characteristic symptom of the disease is the grassy appearance of the affected stools. Numerous thin, small grassy shoots give the plant a bushy appearance. Growth is arrested and the plants remain stunted. When most of the tillers are affected, the entire stool dries up.

Causal Agent

The grassy shoot disease of sugarcane is caused by Sugar Cane Grassy Shoot Virus. This virus is mechanically transmissible and aphid-transmitted. The aphid responsible for transmission is *Aphis sacchari*. The disease is transmitted from sugarcane to sorghum and from sorghum to sugarcane by this aphid.

Control Measures

Seed selection: the seed setts should be selected from healthy canes.

Sanitation: the diseased plants should be dug out and burnt.

Spraying: insecticides should be sprayed to kill the aphids responsible for transmission of the virus.

Hot water treatment: the seed setts should be kept in hot water at 50°C for two hours, prior to sowing.

Avoid sowing of susceptible varieties: the susceptible varieties to this disease should not be sown, for example, Co. 419.

Red Rot of Sugarcane

The pathogen is *Colletotrichum falcatum* Wint., class Deuteromycetes, order Melanconiales, family Melanconiaceae. The perfect stage is known as *Physalospora tucumanensis* Spegazzini, class Ascomycetes, order Sphaeriales, family Mycosphaerellaceae.

Distribution

This disease is undoubtedly the most serious and destructive in India. It is of common occurrence in all countries where sugarcane is grown on a large scale. This disease is found in all the sugarcane growing states of the country. It occurred in a destructive form in eastern Uttar Pradesh and northern Bihar between 1939 and 1942, and in these years, the very existence of the sugar industry was threatened.

Symptoms

The first external symptom of the disease is that the upper leaves of a shoot begin to lose colour and droop slightly. Thereafter they wither at the tip and withering progresses down the margins, leaving the centre green.

All the parts above the ground are infected by the fungus, but the symptoms are conspicuously seen on the stem and midribs of leaves. The infected stems are completely rotted within; the rind loses its naturally bright colour, becomes dull in appearance, and shrinks at the nodes. Black dot-like structures, which are the acervuli of the fungus, appear on the shrunken internodes. On splitting open a cane, an alcoholic smell is emitted, and the tissues are found to be reddened in one or a few internodes, usually towards the base. The reddening is most intense in the vascular bundles but extends to the pith. Often definite red blotches with a white centre, transversely elongated, are found, which are quite characteristic of the disease. Sometimes the reddening is confined to one or a few internodes in the centre of the cane, and the upper and lower parts remain free from disease. As the disease progresses the pith gradually dries up and changes to brown colour. The longitudinal depressions are formed on the rind and canes become very light and easily broken. Cavities develop in the pith, usually containing a dense growth of mycelium. When the canes are dried, the black dot-like acervuli are formed on the rind. The acervuli are formed just above or below the nodes, or in the sunken areas on the internodes.

Conspicuous symptoms also appear on the mid-rib of leaves. A dark-reddish area develops on the mid-rib; it soon elongates and a blood-red lesion with dark margins develops. In mature lesions the centre becomes straw-coloured, and black dot-like acervuli develop in it.

The Pathogen

The red rot of sugarcane is caused by *Colletotrichum falcatum* Wint. The perfect stage is known as *Phyalospora tucumanensis* Spegazzini, of class Ascomycetes, order Sphaerales, family Mycosphaerellaceae. The mycelium of the fungus is inter-cellular and chiefly formed in the parenchymatous cells of the pith. The hyphae are thin, hyaline, septate, profusely branched, and contain oil globules. The acervuli, developed on the rind as well as on the mid-rib of leaves, are minute dot-like, black, and velvety. The long, septate, rigid, bristle-like setae arise from the hymenium of the acervulus; they are up to 200 microns long and 4 microns broad. Numerous clavate, single-celled hyaline conidiophores also arise from this hymenial layer; each conidiophore is 20 microns long and 8 microns broad. The conidia, which develop singly on the conidiophores, are single-celled, hyaline, falcate, thin-walled, granular, and measure 16 to 48 by 4 to 8 microns. In artificial cultures, terminal or intercalary, thick-walled, greenish-black chlamydospores are formed. On germination, the conidia produce single germ tubes at the tips of which the ovoid appressoria are formed.

Water-logging conditions always favour disease incidence. Sometimes rains and humidity also favour the disease.

Nature and Recurrence of Disease

The disease is seed-borne as well as soil-borne. It has been observed that the setts harbour the fungus and thus the disease is perpetuated from year to year. It has also been observed that the fungus is capable of

growing and producing acervuli in soil, but the primary infection is thought to be mainly from infected setts. Secondary spread of disease takes place by means of conidia from the acervuli found on the mid-rib lesions. When the conidia settle on the leaves they germinate and invade the leaves through bruises and wounds. Stem infection takes place through insect borers and root-primordia.

Control Measures

Inspection: the crop should be thoroughly inspected several times, and the infected plants should be destroyed by burning them. All the plants of a single stump should be destroyed.

Avoid water-logging: the water-logging of the field should always be avoided, so that inoculum brought through the water may not infect the healthy plants.

Inspection of setts: prior to sowing, the setts should be duly inspected. If any type of reddening is found, the sett should be discarded for sowing.

Sanitation: sanitation practices should be adopted. After harvesting the crop, the stubbles and plant debris should be carefully destroyed by burning them.

Crop-rotation: crop rotation of two years should be practiced, so that the inoculum present in the soil may be destroyed completely.

Discourage ratooning: the ratooning of infected plants should always be discouraged.

Avoid sowing of susceptible varieties: Co. 312, Co. 331, Co. 213, Co. 290, Co. 210, Co. 313, and Co. 513 have been proved susceptible for the disease. Whenever such varieties were sown, they succumbed; in the U.P. epidemic of 1939 to 1942, the Co. 210 and Co. 213 succumbed.

Resistant varieties: only resistant varieties should be grown. Co. 453, Co. 393, Co. 356, and P.O.J. 7878 were found to be fairly resistant.

Ring Spot of Sugarcane

The pathogen is *Leptosphaeria sacchari* Breda de Haan, class Ascomycetes, order Pseudosphaeriales, family Pseudosphaeriaceae.

Distribution

This is a common disease of sugarcane in several parts of India, especially Uttar Pradesh, Tamil Nadu, West Bengal, Assam, and Maharashtra. This disease was first described in Java, and occurs in most other sugarcane growing countries such as the Philippines, the West Indies, and Australia.

Symptoms

The disease symptoms appear soon after the rains have commenced, when the plants are two to three months old. The first symptom is the appearance of small, discoloured, generally purple spots seen on both surfaces of the leaf. As these spots grow and expand at the margins, the central portion becomes dry. The margin consists of a narrow, reddish purple or brownish band, outside which there is sometimes a yellow areola, merging into the green of the leaf. The centre of the spot is dry and straw-coloured, and sharply marked off by the surrounding coloured ring. Small dot-like structures representing fructifications of the parasite are seen in rows on the straw-coloured spots. The spots are scattered over the whole or part of the leaf-blade, and when fully grown may measure 15 mm by 5 mm. The spots increase in size by the coalescence of neighbouring spots. Ultimately the leaf dries up and withers prematurely.

The Pathogen

The ring spot of sugarcane is caused by *Leptosphaeria sacchari* Breda de Haan. The mycelium is found within the leaf tissues. The hyphae enter the host cells and kill them, making the contents collect in a brown mass in the centre. Perithecia are formed in large numbers, chiefly within the rings; they are arranged in rows as black dots, between the finer veins of the leaf, and remain buried in the leaf tissues, except at the tip. The perithecia are spherical and about 140 microns in diameter. Each perithecium contains numerous slender, elongated, club-shaped asci, mingled with fine paraphyses. Each ascus contains eight ascospores, arranged in 2 or 3 rows. The ascospores are elongated, 3-septate, and measure 20 to 24 by 5 microns.

Predisposing Factors

Unfavourable conditions of soil and moisture favour the disease. The disease is seen on thick canes; thin canes are never damaged.

Nature and Recurrence of Disease

This is a soil-borne disease. The perithecia survive in the soil along with plant debris.

Control

Sanitation is of some use.

Wilt of Sugarcane

The pathogen is *Cephalosporium sacchari* Butler, class Deuteromycetes, order Moniliales, family Moniliaceae.

Distribution

In India, this disease was first recorded by Butler and Khan in 1913. This disease is of common occurrence in Bihar and eastern districts of Uttar Pradesh. It is also known in other countries such as the West Indies, United States, Mexico, and African countries.

Symptoms

This disease produces distinct reddening of the cane pith, and is sometimes confused with red rot disease. On splitting the cane, the pith is found to have a diffused purple or dirty-red colour, which is provided with streaks, but never with the transversely elongated white patches characteristic of red rot. The colour is also not as bright as in red rot. In this disease the pith of the cane dries up more rapidly than when attacked by red rot. The pith becomes hollow and contains fluffy, greyish mycelium.

When the cane is half grown, stunted clumps or even patches are seen in the field. Gradually the withering takes place. The leaves dry up, followed by the stems, which become light and hollow.

The Pathogen

The wilt of sugarcane is caused by *Cephalosporium sacchari* Butler. The mycelium remains confined throughout the reddened portion of the stem. The hyphae penetrate all the cells, bundles, and parenchyma. The conidia are formed on short lateral branches in the vessels. The conidia are hyaline, oval to elliptical, occasionally curved or with one side flattened, measuring 4 to 12 by 2 to 3 microns. The conidiophores are of a definite shape, slightly swollen in the middle, with a blunt end. The conidia can germinate within 24 hours, producing a single terminal germ tube.

Nature and Recurrence of Disease

This disease is both soil- as well as seed-borne. The disease is transmitted from one place to another by means of infected setts. The infected setts make the primary source of inoculum. When diseased setts are

planted, either the eyes do not develop or the shoots arising from the eyes become wilted. Such setts produce a poor root system. The saprophytic fungus can survive in the soil for 2 to 3 years. Alkaline soil favours the disease.

Control

Seeds should be selected from a healthy crop. Ratooning should be discouraged. The crop should not be grown in alkaline soils. Resistant varieties should be sown. Prior to sowing, setts should be dipped in 40 ppm of Boron or Manganese.

Sugarcane Mosaic

The causal agent is Sugarcane Virus Brandes.

Distribution

In India, this disease has been recorded from all the sugarcane growing regions. It is believed that sugarcane mosaic virus originated in the East and has been distributed from there in sugarcane cuttings to all the sugarcane growing countries of the world.

Symptoms

At first pale patches or blotches appear in the green tissues of the leaves. These blotches are uniformly green throughout, but at the same time several shades of lighter colour surround them. They vary in shape and size; usually they are oval or oblong in outline, and their longitudinal axes remain parallel to the mid-rib. The characteristic symptoms are more distinct on the young unrolling leaves; thus symptoms are more distinct on newly opened leaves.

In certain other cane varieties, mottling of the leaves appears; however, this mottling is not shown by all varieties. The entire plants become stunted and chlorotic, and may easily be identified from a distance. In India, the losses caused by this disease are 10 to 20 per cent.

Causal Agent

The sugarcane mosaic is caused by Sugarcane Mosaic Virus Brandes (1919). The virus is transmissible by mechanical means. Several vectors of aphid species have been reported; these vectors are *Rhopalosiphum maidis* Fitch and *Aphis gossypii*, among others. Sugarcane mosaic virus can be partially purified. Purified virus particles are rods measuring 650 to 770 by 12 to 15 millimicrons.

Nature and Recurrence of Disease

The disease is seed-borne. It infects readily by sowing infected seed setts. Once the virus enters the host plant, it becomes systemic.

Control

Resistant varieties should be grown. Systematic rogueing of the infected canes should be done. Seed setts should be selected from a healthy crop. The wild hosts, such as Johnson grass (*Sorghum halepense*), should be eliminated. Quarantine regulations should be strictly followed.